JJJ-O-533E
August 27, 1971
SUPERSEDING
Fed. Spec. JJJ-O-533D
February 16, 1967

FEDERAL SPECIFICATION

ONIONS, DEHYDRATED

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers chopped, powdered, sliced, minced and granulated dehydrated white onions, packaged for domestic and overseas shipment and intended for use by agencies of the Federal Government.

1.2 Classification. The dehydrated onions covered by this specification shall be of the following styles, and as specified (see 6.1):

Styles:

1 - Chopped
2 - Powdered
3 - Sliced
4 - Minced
5 - Granulated

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans.
PPP-B-636	-	Box, Fiberboard.
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment **(Civil Agencies).**

FSC 8915

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(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items.
MIL-L-35078 - Loads, Unit: Preparation of Nonperishable Subsistence.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-668 - Sanitary Standards for Food Plants.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Government:

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

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(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

United States Standards for Condition of Food Containers.

(Application for copies should be addressed to the Information Division, Consumer and Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

Nongovernmental:

Association of Official Analytical Chemists:

Official Methods of Analysis of the Association of Official Analytical Chemists.

(Copies may be obtained upon application to the Association of Official Analytical Chemists, P. O. Box 540, Benjamin Franklin Station, Washington, DC 20044.)

National Motor Freight Traffic Association, Inc. Agent:

National Motor Freight Classification.

(Application for copies should be addressed to American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N. W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 202, Union Station, 516 W. Jackson Blvd., Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. The onions used for dehydration shall be clean, sound, and wholesome, and shall be of the white varieties, having pungent characteristics and a solids content of not less than 10 percent. Varieties which yield a bitter or mild dehydrated product shall not be used. The onions shall be free from soft rot, and from damage caused by dirt, or other foreign matter, seed stems, moisture, sunburn, sunscald, disease, insects, or mechanical or other means.

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3.2 Preprocessing. The onions shall be washed thoroughly to remove soil and sorted to remove any which are diseased or otherwise damaged. The root bases (see figure 1), tops and outer discolored scales shall be removed before or after dehydration.

3.3 Dehydrating. The dehydration conditions shall be such that the product will not be scorched or otherwise damaged by heat or drying.

3.4 Finished product. The product and packages shall be free from insect infestation and shall comply with 3.4.1 through 3.6. Calcium stearate or other anticaking agents in quantities permissible under the provisions of the Federal Food, Drug, and Cosmetic Act may be added to the style 2 or style 5 product to prevent caking.

3.4.1 Dehydrated product.

3.4.1.1 Moisture content. The moisture content of the packaged dehydrated product shall not exceed 4.0 percent by weight.

3.4.1.2 Particle or piece size. Particle or piece size shall be expressed in percent, by weight.

3.4.1.2.1 Style 1, chopped. Not less than 40 percent shall be retained on a U.S. Standard No. 8 sieve and not more than 10 percent shall pass through a U.S. Standard No. 12 sieve.

3.4.1.2.2 Style 2, powdered. The powdered product shall be free-flowing with not less than 98 percent passing through a U.S. Standard No. 45 sieve, not less than 75 percent passing through a U.S. Standard No. 80 sieve, and not less than 50 percent passing through a U.S. Standard No. 100 sieve.

3.4.1.2.3 Style 3, sliced. Not less than 10 percent shall be retained on a U.S. Standard 1/4-inch sieve, and not more than 5 percent shall pass through a U.S. Standard No. 12 sieve.

3.4.1.2.4 Style 4, minced. Not more than 2 percent shall be retained on a U.S. Standard No. 6 sieve, and not more than 3 percent shall pass through a U.S. Standard No. 20 sieve.

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3.4.1.2.5 Style 5, granulated. Not more than 2 percent shall be retained on a U.S. Standard No. 35 sieve and not more than 5 percent shall pass through a U.S. Standard No. 100 sieve.

3.4.1.3 Hot water insoluble solids. The hot water insoluble solids shall not exceed 20 percent when determined in accordance with 4.3.2.4.

3.4.1.4 Color. The dehydrated onions shall be free of yellow onions when tested in accordance with 4.3.2.6 and shall meet the optical density requirements specified in table I when tested in accordance with 4.3.2.3.

TABLE I. Optical density requirements

Colors	Styles	Maximum optical density
White	1,3, or 4,	0.090
	2 or 5	0.105

3.4.1.5 Defects. Dehydrated onions shall not exceed the defects for the respective styles specified in table II.

TABLE II. Defect tolerances

Styles	Defects	Not more than
2	Acid insoluble ashed sediment	0.04%, by weight
4	Dark brown or black specks	20 specks in a 25 gm sample
5	Dark brown or black specks	20 specks in a 0.1 gm sample
1 and 3	Defective pieces	0.5%, by weight

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Defective pieces are dark brown or black pieces, seed stems, outer skin (tissue-like membrane of the outer surface of the onion bulb), outer root base or roots as illustrated herein, sediment and extraneous vegetable material. The defective pieces shall be separated from sound onion before determining their weight.

3.4.2 Rehydrated product.

3.4.2.1 Rehydration.

3.4.2.1.1 Styles 1,3 and 4 rehydration procedure. The onions shall be rehydrated by adding 4 ounces of dehydrated product to 24 fluid ounces of water at 75 ± 5° F. and allowing to soak for one hour.

3.4.2.1.2 Style 2 rehydration procedure. Style 2 shall be rehydrated by adding 1 ounce to 2 fluid ounces of water at 75° F. ± 5° F.; while stirring.

3.4.2.1.3 Style 5 rehydration procedure. Style 5 shall be rehydrated by adding 1 ounce to 4 fluid ounces of water at 75° F. ± 5° F.; while stirring.

3.4.2.2 Flavor. Dehydrated onions rehydrated in accordance with 3.4.2.1.1, 3.4.2.1.2 or 3.4.2.1.3, as applicable, shall possess a good flavor and odor characteristic, pungent and free from any objectionable flavors or odors.

3.4.2.3 Texture. The rehydrated onions, styles 1, 3 and 4, shall possess a good tender, fleshy texture similar to that of fresh onions.

3.5 All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

3.6 Workmanship. The product shall be prepared, processed, and packaged under modern sanitary conditions and in accordance with good commercial practice. For military agencies the product shall be processed in establishments meeting the requirements of MIL-STD-668.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling. Samples of the finished products covered by this specification and samples of any materials entering into the preparation of the finished products may be taken at any time by the Government Inspector to determine compliance with the requirements.

4.2 Inspection.

4.2.1 General. Unless otherwise specified in the contract, inspection shall be made by the procuring agency or a duly authorized representative, at the time and place designated by the procuring agency. It may be made at the site of manufacture, both during and after processing, at any suitable point in transit, or after delivery at the point of destination. The passing as satisfactory of any detail of processing or materials shall not relieve the supplier of responsibility for faulty workmanship or materials which may be discovered at any time prior to final acceptance.

4.2.2 Final inspection. Unless otherwise specified, final inspection shall be made after delivery to the point of destination (see 6.4).

4.2.3 Condition inspection of containers. When inspection of product for quality and condition is performed by the U.S. Department of Agriculture, the United States Standards for Condition of Food Containers shall apply. (For applicable criteria governing inspection for the military see 4.2.6.3.)

4.2.4 Responsibility for inspection. The supplier is responsible for meeting all the provisions of the specification prior to submission of supplies to the Government for acceptance. Unless otherwise specified in the contract or purchase order, the supplier is required to perform the examination and tests prescribed herein, using his own or any other inspection facilities and services acceptable to the Government.

4.2.4.1 Inspection records of the examination and tests shall be complete and made available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2.5 Preacceptance inspection (plant sanitation inspection for military agencies). Product shall be rejected if produced in plants not meeting the requirements of MIL-STD-668.

4.2.6 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.6.1 Component and material inspection. In accordance with 4.2.4, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification, or applicable purchase document.

4.2.6.1.1 Testing of cans and onions. Testing of cans and onions shall be in accordance with table III. Test requirements for the characteristics listed shall be average requirements. A test failure shall indicate an unacceptable component and use of such shall be cause for rejection of the involved end item.

TABLE III. Testing of cans and onions

Component	Sample unit	Lot size expressed in	Inspection level	Characteristics	Results reported to	Test ref and reqmt para
Cans 1/	1 can w/lid	cans	S-1	Tin plate	Nearest 0.01 lb. base box	4.3.2.7 and 5.1.1.3
Fresh onions	1-pound Composite	pounds of onions	2/	Solids	Nearest percent	3.1 and 4.3.2.1

1/ Testing of cans for can coating (when required and can label shall be performed on the same cans submitted for tin plate test.
2/ The sample shall be randomly selected from the lot of fresh onions.

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4.2.6.1.2 Envelopes. Conformance of envelopes to the requirements of 5.1.1.1 and 5.1.1.2, as concerns thickness of each lamina and basis weight of paper shall be ascertained by examination of pertinent invoices or other valid documents from the supplier of envelope material. Nonconformance with requirements shall indicate an unacceptable component and use of such shall be cause for rejection of the involved end item.

4.2.6.2 Process examination. Unless otherwise specified (see 6.1) examination shall be performed to determine conformance to 3.2. Nonconformance to required procedures shall be cause for rejection of the involved finished product.

4.2.6.3 Examination of end item. Examinations shall be in accordance with tables V through VIII and 4.2.6.3.1 through 4.2.6.3.4. The acceptable quality levels (AQLs) shall be expressed as percent defective for tables V and VI and defects per hundred units for tables VII and VIII. Sampling plans shall be in accordance with table IV.

TABLE IV. Sampling plans

Table	Sample unit	Lot size expressed in	Inspection level	AQLs	
				Major	Minor
V	1 filled primary container	Primary containers	S-3	--	2.5
VI	1 bulk packed shipping container with envelopes or 1 filled carton from a shipping container	Shipping containers	S-2	--	4.0
VII	1 filled and sealed envelope or can	Envelope or can	S-3	0.65	--
VIII	1 filled primary container	Primary container	S-2	1.5	--

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TABLE V. Examination for net weight 1/

Category	Defect
Minor	
201	Net weight more than 5 percent under specified weight (container required to have less than 16 ounces). 2/
202	Net weight more than 2 percent under specified weight (containers required to have 16 ounces or more). 3/

1/ Lot shall be rejected if sample data indicate a lot average net weight less than specified net weight.
2/ Report results to nearest 0.01 gram.
3/ Report results to nearest 1/10 ounce.

TABLE VI. Examination for count of envelopes per container 1/

Category	Defect
Minor	
201	Less than 975 envelopes per carton or bulk packed shipping container.

1/ Lot shall be rejected if sample data indicate a lot average count less than 1000 envelopes.

TABLE VII. Examination for leakage and delamination of envelopes (see 4.3.1.4 for procedure)

Category	Defect
Major	
151	Leakage.
152	Delamination. 1/

1/ Examine envelope after removal from water.

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TABLE VIII. Examination of product characteristics 1/

Category	Defect
Major	
	Dehydrated
101	Style 4 product has more than 20 dark brown or black specks in 25 gm 2/
102	Style 5 product has more than 20 dark brown or black specks in 0.1 gm 2/
103	Styles 1 or 3, has more than 0.5% by weight of defective pieces as specified in table II. 2/
104	Particle or piece size (except style 1 in envelope and style 2) not as specified 2/
	Rehydrated (see 3.4.2.1 for procedure)
105	Styles 1, 3 or 4 do not rehydrate essentially to its fresh form
106	Style 2 does not rehydrate to a smooth paste
107	Style 5 does not rehydrate to a uniformly particulate paste
108	Flavor and odor not similar to good fresh onions
109	Objectionable flavor or odor
110	Texture of styles 1, 3 or 4, product not similar to fresh onion texture

1/ Presence of foreign material (e.g. glass, wood, metal, paint, insects, filth, etc.) shall be cause for rejection of the lot.

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2/ See 4.3.1.2 or 4.3.1.3 as applicable for procedure. Make one determination per sample unit.

4.2.6.3.1 Examination of particle size for style 1 product in envelopes and style 2 product in cans and envelopes. The product shall be examined for particle size in accordance with 4.3.1.3. The sample for style 2 product in cans shall be a 100-gram composite derived from the number of cans indicated by inspection level S-2, or for envelopes, the number of envelopes necessary to yield 100 grams. Lot size shall be expressed in primary containers. Non-conformance to requirements shall be cause for rejection of the lot.

4.2.6.3.2 Examination of primary containers for external condition and labeling. Examinations shall be in accordance with the criteria in the United States Standards for Condition of Food Containers.

4.2.6.3.3 Examination of shipping containers. Examination of filled and closed shipping containers shall be in accordance with the quality assurance provisions in the appendix of PPP-B-636. In addition, the following defects shall be included in the examination: Major - markings missing, incorrect or illegible. Minor - arrangement of cans not as specified. For level C pack as specified in 5.2.3.1 only the defects applicable to marking and arrangement of cans shall apply.

4.2.6.3.4 Examination of unit loads. Unit loads shall be examined in accordance with the quality assurance provisions of MIL-L-35078.

4.2.7 Testing of finished product. The finished product shall be tested for moisture, hot water insoluble solids and optical density and acid insoluble ash for style 2 product only and adulteration by yellow onions in accordance with 4.3. Test requirements shall be lot average requirements. The sample for testing shall be a 1-pound composite derived from the number of cans indicated by inspection level S-2 or the number of envelopes to yield 1 pound. Lot size shall be expressed in terms of primary containers. Each result except adulteration by yellow onions shall be reported to the same decimal place as specified in the pertinent requirement. Result of test for adulteration by yellow onions shall be reported as pass or fail. Test failure shall be cause for rejection of the lot.

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4.3 Test methods. Unless otherwise excepted, examination and test procedures which differ from those specified herein, may be used by the supplier if they provide a quality assurance equivalent to that specified. If the Government contracting officer determines that such procedures and controls do not provide, as a minimum such quality assurance, the supplier will use the procedures set forth herein. In case of dispute as to examination or test results, the procedures specified herein will govern.

4.3.1 Examination procedures.

4.3.1.1 Examination for defects, styles 1 and 3. Styles 1 and 3 shall be examined for defective pieces by weighing 100 grams, spreading the sample on a white cardboard sheet, carefully picking out the defective pieces and determining the percentage of the latter by weight. Defective portions shall be broken from the sound onion pieces prior to weighing.

4.3.1.2 Procedure for speck count, styles 4 and 5.

Equipment and materials

Wide view (20-30X) dissecting scope.
Balance accurate to 0.10 gm.
Petri dish with ruled filter paper attached underneath for granulated onions.
Enameled pan for minced products.

Procedure

1. Weigh out 0.1 gm granulated onion or 25 gm minced onion.
2. Transfer to petri dish or enamel tray.
3. Spread uniformly.
4. Count dark brown or black specks. Green pigmented pieces are not considered specks.

4.3.1.3 Particle or piece size screening. Particle size screening shall be performed as follows: spread approximately 100 g. on 8-inch diameter or 200 g on 12-inch diameter sieves. Shake the sieves in a Rotap with tappet operating, or an equivalent shaker, for 2 minutes. Remove onions retained on sieve, or sieves, and repeat procedure until the total sample or contents of the can have been screened. The total of the material retained and passing through the sieve or sieves, shall be weighed and used in the determination of compliance with 3.4.1.2. When calcium stearate or other anticaking agents have been added, the amount added should be subtracted from the calculations.

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4.3.1.4 Leakage of envelopes and cans. Determination of leakage shall be performed as follows: submerge the envelope or can to be examined in water in a vacuum desiccator, Mead tester, or equivalent device. Draw 10 inches of vacuum and hold for at least 10 seconds. Observe envelope or can, for (a) air leakage in quantities from any point, (b) for a fine stream of air bubbles anywhere in the seal or seam area, and (c) for failure of the envelope to inflate. Envelopes which do not inflate under vacuum should be opened and examined for water penetration. Leakage of air under the first two conditions, or wetting of the product under the last condition, indicates a defect.

4.3.2 Test procedures.

4.3.2.1 Solids content of the onions used for dehydration. The solids content of the onions to be used for dehydration shall be determined in accordance with the preparation of samples and total Solids methods under Canned Products in the chapter for Vegetable Products Processed of the Official Methods of Analysis of the Association of Official Analytical Chemists.

4.3.2.2 Moisture determination. Moisture shall be determined as follows:

(a) Preparation of sample for styles 1, 3 and 4. Grind approximately 100 g. of a well mixed sample in a Wiley, or similar type mill, equipped with a 20-mesh screen under low humidity conditions. Style 2, powdered, and style 5, granulated, onions need no further preparation.

(b) Procedure. Weigh two approximately 2 g. samples of ground material into tared, dry, aluminum weighing dishes (approximately 2 or 2-1/2 inches in diameter and 3/4 inch in depth) equipped with tight-fitting covers. Place the dishes with cocked lids in a vacuum oven and dry for 16 hours at 60°C. under a pressure of not more than 100 millimeter (mm.) mercury. During drying, admit to the oven a slow current of air (approximately 2 bubbles per second) dried by passing through concentrated sulfuric acid. Remove the dishes and allow them to cool in a desiccator before weighing. Calculate the percent of moisture in the original samples.

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4.3.2.3 Optical density.

Equipment and materials.

Spectrophotometer.
2 cells.
Wide mouth pint jars.
Ribbed, stemless glass funnel of 4-1/4 inches outside diameter.
Triple beam balance.
Whatman #12 fluted filter paper (18.5 cms.).
100 ml burette.
10 percent solution of sodium chloride (reagent grade).

Procedure.

1. Weigh 2.0 grams of ground onion product (see 4.3.2.2) and transfer to a wide mouth pint jar. (Depending on the equipment used for measuring optical density, sample weight can be varied so that the readings obtained fall into the optimum range for the instrument. If sample size is varied, corrections must be made for this in the final calculations).

2. Measure out 100 ml of salt solution, add a small amount of it to the sample and make a smooth slurry.

3. Gradually add the rest of the salt solution.

4. Allow the suspension to stand 15 minutes with occasional stirring.

5. Filter through Whatman #12 filter paper, re-filter until crystal clear.

6. Determine percent transmission of filtrate at 420 millimicron wave length with a Colorimeter or a Spectrophotometer standardized at 100 percent transmission with the 10 percent salt solution used for extraction. When instruments with adjustable light transmission slits are used, the narrowest slit width shall be used for greater precision.

7. Calculate optical density on the basis of a 1.0 percent solution and 50 mm. cell path.

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4.3.2.4 Acid insoluble ashed sediment.

Equipment and materials

Separatory funnel-500 ml.
Evaporating dishes-preferably platinum, 50-75 ml capacity. Clean dishes should be heated in muffle furnace and cooled in desiccator.
Muffle oven.
Drying oven.
Steam bath.
Desiccator.
Filter paper-Whatman #40.
pH paper-pH 7.
Analytical balance (0.1 mg precision).
Carbon tetrachloride.
Hydrochloric acid solution (1 part conc. HCl+2-1/2 parts distilled water).

Precaution: pour acid slowly into water.

Procedure

1. Suspend 50 gm of powdered onion into 200 ml of carbon tetrachloride in a 500 ml separatory funnel. Mix well and let stand for a prolonged period-preferably overnight.

2. Draw out the sediment from the separatory funnel into a tared evaporating dish. Flush the remaining sediment from the separatory funnel with an additional 30-40 ml of carbon tetrachloride.

3. Evaporate slowly to dryness on a hot plate in a hood. CAUTION: Carbon tetrachloride fumes are extremely toxic and evaporations should only be done in a well ventilated hood.

4. Place dish in muffle oven at 550°C. until ash is white. Cool in desiccator.

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5. To the ash in evaporating dish add 25 ml HCl solution and boil slowly on steam bath for 5 minutes; keep dish covered with a watch glass to prevent splattering.

6. Filter mixture (hot) through a Whatman #40 Ashless filter paper, washing dish and residue with hot water until acid free. Use pH paper to ascertain whether acid free.

7. Place filter paper containing residue into original evaporating dish. Dry for one (1) hour at 100°C. in drying oven.

8. Ash in muffle oven at 550°C. for 2 to 3 hours or until ash is white. Then allow to cool in desiccator.

9. Weigh dish and contents.

10. Calculations:

Percent AIAS=$2(D_C-D_T-F_A)$.

Where:

D_C=Dish plus contents.
D_T=Tare weight of dish.
F_A=Filter paper ash (given on filter paper box).

4.3.2.5 Hot water insoluble solids (HWI).

Equipment and materials

Hot plate
Tall graduated 400 ml beakers.
Filter aid.
Drying oven operating at 105°C.
Analytical balance (1.0 mg precision).
Vacuum system and suction flask.
Anti-foam spray.
Gooch crucible.
Filter paper: Whatman #540 to fit crucibles.
Wash bottle with distilled water near boiling.
Stirring rods with rubber policeman.
Desiccator.

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Procedure

1. Prepare a clean, constant weight Gooch crucible containing a disc of Whatman #540 ashless filter paper.

2. Suspend 1.500 gm of ground onion (see 4.3.2.2)(a) in 200 ml of distilled water in a tall 400 ml beaker.

3. Add 1.500 gm of dried filter aid and stir.

4. Heat to boiling and boil slowly for 5 minutes.

5. Using the vacuum suction system, filter through tared Gooch crucible and wash with 200 ml distilled water near boiling.

6. Dry to constant weight at 105°C. Cool in desiccator before weighing.

7. Repeat a blank determination with 1.500 gm filter aid only.

8. Calculations:

$$\text{Percent HWI} = \frac{C_R - C_T - F_B}{S_W} \times 100$$

Where:

C_R=Weight of crucible plus residue
C_T=Tare weight of crucible with filter paper
F_B=Weight of filter aid blank residue
S_W=Sample weight

4.3.2.6 Adulteration by yellow onions. Adulteration by yellow onions shall be determined as follows:

Procedure:

1. Thoroughly mix 1 g. of the ground onion product (see 4.3.2.2)(a) with 10 ml. of distilled water.

2. Add 1 ml. of 5 percent sodium hydroxide and mix thoroughly.

3. The formation of a bright yellow color indicates the presence of yellow onion.

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4.3.2.7 Tin coating weights. Tin coating weights shall be determined by any method specified in PPP-C-96.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The dehydrated onions shall be packaged in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. The dehydrated onions shall be packaged in accordance with 5.1.1.1 through 5.1.1.3, as applicable.

5.1.1.1 Style 1 (envelopes). One-quarter ounce or 1/2-ounce of style 1 dehydrated onions, as specified (see 6.1), shall be packaged in a heat-sealable envelope having maximum outside dimensions of 4 by 4 inches (1/4-ounce envelope) or 4 by 5 inches (1/2-ounce envelope). All seams and closures shall be made by heat sealing. The envelope material shall be constructed by hot-extruding polyethylene to 0.00035-inch thick aluminum foil and laminating the plain side of the foil to sulphite or Kraft paper. The polyethylene shall have a minimum thickness of 0.001 inch. The paper shall have a basis weight of 40 pounds per ream ± 5 percent (24 by 36-500) and have a white or natural color. The laminant shall be an odorless and permanently plastic water resistant material other than wax. The combined sheet shall show no evidence of delamination when fabricated into envelopes and heat-sealed. The filled and sealed envelope shall not show leakage or delamination when tested in accordance with 4.3.1.4.

5.1.1.2 Style 2, and 5 (envelopes). Two and one half grams of style 2 or 5 dehydrated onions, shall be packaged in a heat-sealable envelope having maximum outside dimensions of 2-3/8 by 2-5/8 inches. All seams and closures shall be made by heat sealing. The envelope material shall be the same as that specified in 5.1.1.1, except that the basis weight of the sulphite or Kraft paper shall be 25 pounds per ream (24 by 36-500) and shall be treated to form a soft pliable sheet. The envelope material shall be tested in accordance with the tests specified in 5.1.1.1.

5.1.1.3 Styles 1, 2, 3, 4 and 5 (cans). Dehydrated onions of the style specified (see 6.1) shall be filled in the can size specified (see 6.1), with the net weight specified for the respective styles in table IX. The can shall be an open-top style, round, hermetically sealed metal can with soldered side seam and compound-lined, double-seamed ends, made throughout from not less than commercial 0.25 pound electrolytic tin plate per base box (see 4.3.2.7). The can shall

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be coated overall on the outside with a coating conforming to type I or when specified type III of TT-C-495 (see 6.1). The filled and sealed can shall not leak when examined in accordance with 4.3.1.4.

TABLE IX

Style, can size, and fill weight			
Style	Weight (pounds, ounces)		Can size
1	2	8	603 x 700
2	1	-	401 x 411
2	4	-	603 x 700
3	1	12	603 x 700
4	3	-	603 x 700
5	1	-	401 x 411
5	4	-	603 x 700

5.1.2 Level C. All styles, cans only. The product shall be packaged as specified in 5.1.1.3 except that cans with or without a commercial exterior coating will be acceptable.

5.1.2.1 **For Veterans hospitals use only.** When specified (see 6.1), one pound and twelve ounces of the product shall be packaged in a foil-laminated bag, with a heat sealed closure.

5.2 Packing, level A, B and C. The product shall be packed in snug-fitting shipping containers in accordance with level A, B, or C, as specified (see 6.1) and as specified below.

(a) Twenty-four size 401 by 411 cans, arranged 4 in length, 3 in width and 2 in depth, or six size 603 by 700 cans, arranged 3 in length, 2 in width, and 1 in depth, shall be packed on end in the shipping container.

(b) Envelopes (style 1). One thousand of the 1/4-ounce or 1/2-ounce envelopes, packaged as specified in 5.1.1.1, shall be random packed in the shipping container specified in 5.2.3.1.

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(c) Envelopes (style 2 or 5). One thousand of the 2-1/2 gram envelopes packaged as specified in 5.1.1.2 shall be packed into an intermediate folding or set-up carton. The intermediate cartons shall be overpacked in the shipping container specified in 5.2.3.1.

5.2.1 Level A. The shipping container shall be constructed, closed and reinforced in accordance with style RSC-SL,V2s, of PPP-B-636. Sleeve and reinforcement are not required when unit loading is specified.

5.2.2 Level B. The shipping container shall be constructed and closed in accordance with style RSC, W5c or W5s, of PPP-B-636.

5.2.2.1 When specified (see 6.1 and 6.6) the shipping container specified in 5.2.2 shall be reinforced with pressure-sensitive adhesive, filament-reinforced tape.

5.2.3 Level C. The shipping container shall be constructed and closed in accordance with style RSC, type CF or SF, class domestic, method II closure, of PPP-B-636.

5.2.3.1 When specified (see 6.1 and 6.7) the product packaged in cans, bags or envelopes shall be packed in a shipping container complying with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Unit loads. When specified (see 6.1), the product packed as specified in 5.2 shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified.

5.4 Labeling and marking.

5.4.1 Civil agencies.

5.4.1.1 Unit containers. Unless additional labeling is specified, any labeling that complies with the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder will be acceptable.

5.4.1.2 Shipping containers. Unless otherwise specified, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.4.2 Military agencies.

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5.4.2.1 Cans. Cans shall be plainly marked in accordance with MIL-L-1497, as follows:

ONIONS, DEHYDRATED (Chopped, Sliced or Minced as applicable)

NET WEIGHT

NAME AND ADDRESS OF PACKER

DIRECTION FOR USE

1. For each pound dehydrated onions, add 3 qts. water at room temperature (70°F. to 90°F.).

2. Let stand 30 minutes. Drain. Use as raw onions.

3. If recipe calls for the addition of liquid, save rehydration water.

NOTE: One pound of dehydrated onions is equivalent to 10 pounds of unpeeled fresh onions. If storage is necessary after opening, use moisture-proof containers under refrigeration.

ONIONS, DEHYDRATED (Powdered or Granulated, as applicable)

NET WEIGHT

NAME AND ADDRESS OF PACKER

USE FOR SEASONING

NOTE: One pound of dehydrated onions is equivalent to 10 pounds of unpeeled fresh onions. If storage is necessary after opening, use moisture-proof containers under refrigeration.

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5.4.2.2 Envelopes. The following information shall be printed on the face of the envelope in addition to any other information required by the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

ONIONS DEHYDRATED (in letters larger than any others used for the label)

STYLE (as applicable)

NET WEIGHT

USE FOR SEASONING

NAME AND ADDRESS OF PACKER

5.4.2.3 Intermediate and shipping containers. Intermediate and shipping containers shall be marked in accordance with MIL-STD-129.

5.4.2.4 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Style of product required (see 1.2).
(c) When process examination is not required for civil agencies (see 4.2.6.2).
(d) Levels of packaging and packing (see 5.1 and 5.2).
(e) Quantity of style 1 to be packaged per envelope (see 5.1.1.1).
(f) When type III exterior can coating is required for level A shipment (see 5.1.1.3).
(g) Style of onion and can size required (see 5.1.1.3).
(h) When packaging specified in 5.1.2.1 is required.
(i) When packing specified in 5.2.2.1 is required.
(j) When packing specified in 5.2.3.1 is required.
(k) Type and class of unit load required, when specified (see 5.3).

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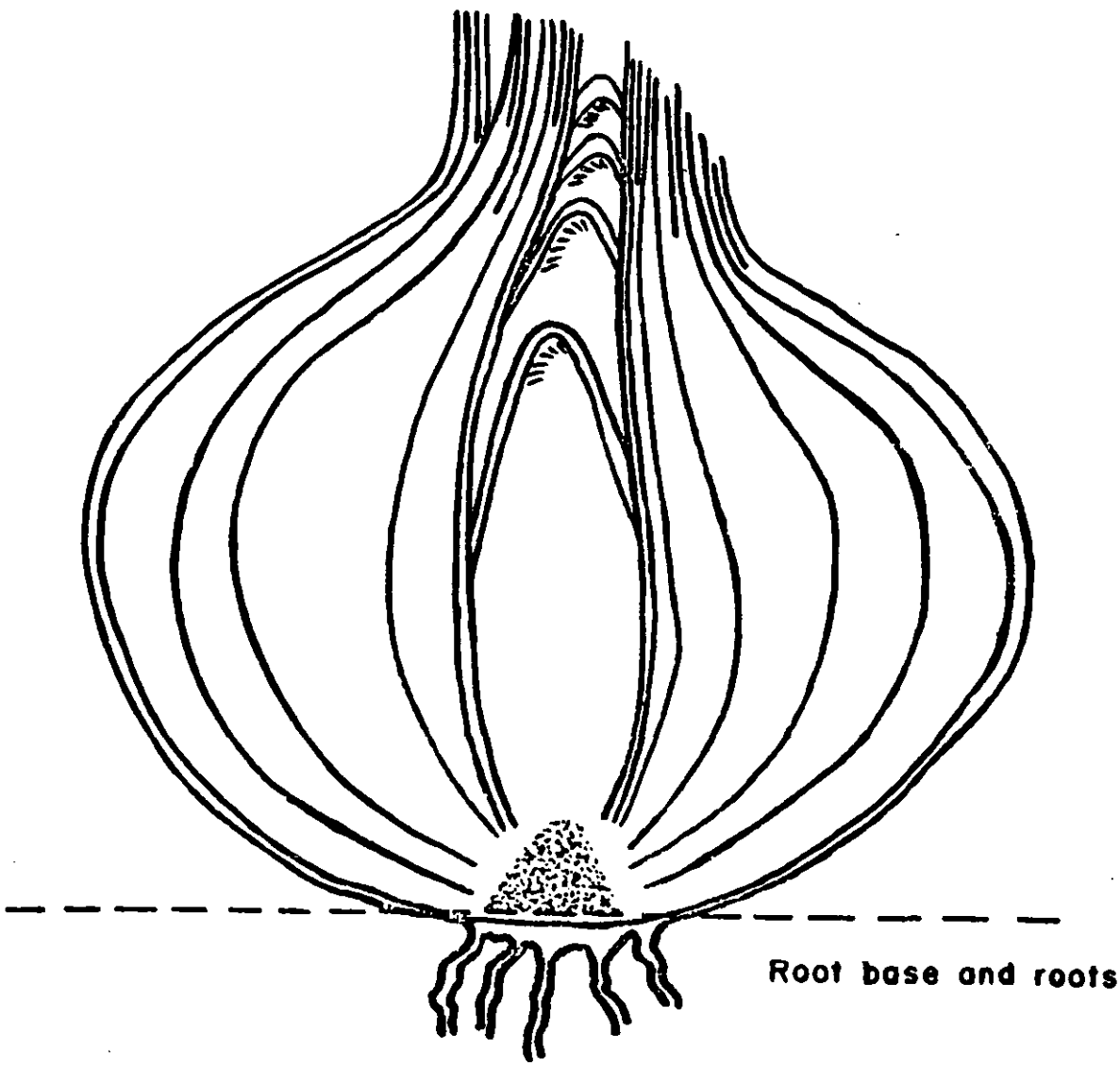


Figure 1. Drawing of onion indicating root base.

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6.2 If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated.

6.3 Supplier's inspection. The supplier should be required to assure that the product conforms with the specification prior to submission to the Government for final inspection.

6.4 Destination inspection (civil agencies only). When the finished product has been inspected and passed at point other than destination, the contract should require that the product be inspected at destination for identity, condition, and quantity only.

6.5 Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the contracting officer should select the appropriate level of pack in accordance with the criteria established in AR 700-15.

6.6 Packing specified in 5.2.2.1 is intended for transfer at sea operations and specific overseas operations.

6.7 Packing specified in 5.2.3.1 is intended for direct shipment of cans from the supply source to the first receiving activity for immediate use within CONUS or for direct shipment of envelopes to a ration assembler.

MILITARY CUSTODIANS:

Army - GL
Navy - SA
Air Force - 45

Review activities:

Army - MD
Navy - MC, MS
DP - SS

Preparing activity:

Army - GL

CIVIL AGENCIES INTEREST:

GSA
AGR
HEW
VA

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

Onions, Dehydrated JJJ-O-533E

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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